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Crowding distance and IGD-driven grey wolf reinforcement learning approach for multi-objective agile earth observation satellite scheduling

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ABSTRACT

With the rise of low-cost launches, miniaturized space technology, and commercialization, the cost of space missions has dropped, leading to a surge in flexible Earth observation satellites. This increased demand for complex and diverse imaging products requires addressing multi-objective optimization in practice. To this end, we propose a multi-objective agile Earth observation satellite scheduling problem (MOAEOSSP) model and introduce a reinforcement learning-based multi-objective grey wolf optimization (RLMOGWO) algorithm. It aims to maximize observation efficiency while minimizing energy consumption. During population initialization, the algorithm uses chaos mapping and opposition-based learning to enhance diversity and global search, reducing the risk of local optima. It integrates Q-learning into an improved multi-objective grey wolf optimization framework, designing state-action combinations that balance exploration and exploitation. Dynamic parameter adjustments guide position updates, boosting adaptability across different optimization stages. Moreover, the algorithm introduces a reward mechanism based on the crowding distance and inverted generational distance (IGD) to maintain Pareto front diversity and distribution, ensuring a strong multi-objective optimization performance. The experimental results show that the algorithm excels at solving the MOAEOSSP, outperforming competing algorithms across several metrics and demonstrating its effectiveness for complex optimization problems.

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1. Introduction

Agile Earth observation satellites (AEOSs) are characterized by their enhanced attitude controllability and ability to perform multiple observation tasks within extended visible time windows. Unlike conventional Earth observation satellites (EOSs), which are limited by fixed observation angles and can only collect data within a constrained range as they follow their orbits (as shown on the left side

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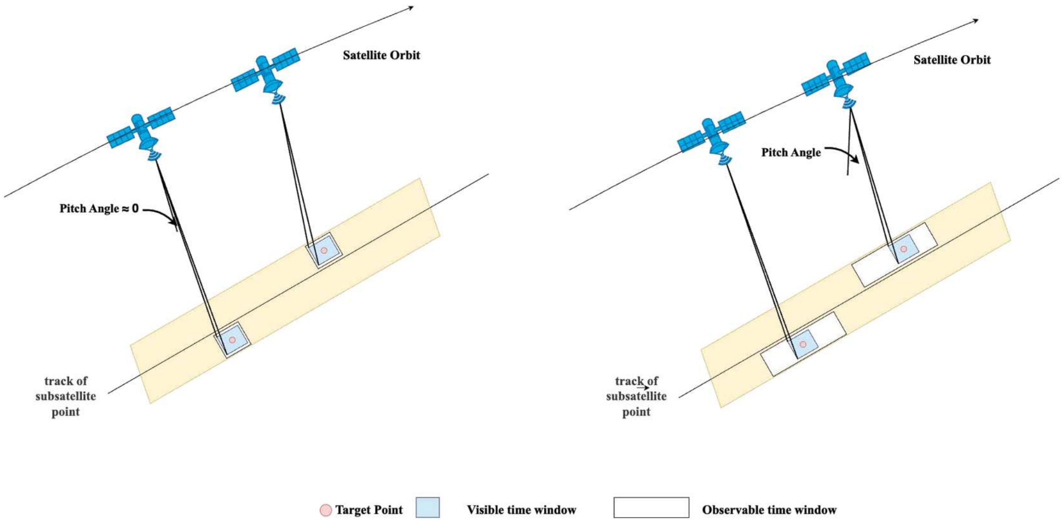


Figure 1. Schematic diagram of the operation of an EOS and an AEOS.

of Figure 1), AEOSs have the ability to adjust their attitude along the roll, pitch, and yaw axes. This flexibility allows AEOSs to increase their pitch angle, enabling a broader observation time window (as shown on the right side of Figure 1). Consequently, AEOSs can observe multiple, widely dispersed targets within a single orbit, significantly enhancing observation efficiency. This flexibility is crucial for meeting the diverse demands of users, leading to extensive applications in fields such as military reconnaissance (Song et al. 2023), disaster response (Niu, Tang, and Wu 2018), environmental monitoring (Hao et al. 2012), and precision agriculture (Song, Ou, et al. 2024). However, as user requirements become more complex, the scheduling of AEOSs faces increased challenges, particularly in balancing diverse user needs while optimizing multiple objectives with constrained resources. The crux of the issue lies in the optimal allocation of resources and the strategic prioritization of tasks to maximize satellite utilization, all while satisfying the varied needs of multiple users and tasks. Research in this area not only facilitates precise responses to complex demands but also supports the broader application of agile satellites across various domains, thereby fully capitalizing on their pivotal role in modern remote sensing technology.

The AEOS scheduling problem (AEOSSP) is characterized by its multi-objective, multi-constraint, and high-dimensional nature; it has been proven to be NP-hard (Lemaître et al. 2002; Wei et al. 2021). Consequently, resource allocation and task scheduling become highly complex. Traditional exact algorithms, such as integer linear programming (a framework that often employs the branch and bound method) (Chu, Chen, and Tan 2017; Spangelo et al. 2015) and dynamic programming (Nag, Li, and Merrick 2018), can solve such problems. However, they often exhibit inefficiency when dealing with large-scale and highly complex scenarios, making it challenging to obtain feasible solutions within a reasonable timeframe. As a result, researchers have increasingly turned to heuristic and metaheuristic algorithms, including genetic algorithms (Chatterjee and Tharmarasa 2022), particle swarm optimization (Gu et al. 2023), ant colony optimization (Yu et al. 2020), simulated annealing (Stollenwerk et al. 2021), and tabu search (Habet, Vasquez, and Vimont 2010); these algorithms have become common choices for finding near-optimal solutions.

Currently, research on the AEOSSP is primarily focused on the optimization of a single objective function. Peng et al. (2022) proposed a mathematical model optimized through a genetic algorithm for satellite observation planning, and validated its effectiveness through computational experiments. L. He et al. (2018) proposed an adaptive large neighborhood search (A-ALNS) algorithm for multi-satellite scheduling, demonstrating superior task completion and robustness compared to existing methods. Z. Zhou et al. (2023) addressed the variable-duration imaging problem for AEOSs using a diminishing marginal returns function and proposed an improved adaptive ant colony optimization (IAACO) algorithm, which outperformed traditional ACO and A-ALNS algorithms in terms of benefits and convergence. Han et al. (2023) used a simulated annealing based heuristic method with a quick insertion algorithm to solve the multi-satellite scheduling problem, maximizing observational benefits. The experimental results demonstrated its effectiveness in finding near-optimal solutions within a reasonable timeframe.

Despite significant advancements in single-objective optimization, the practical applications of the multi-objective agile Earth observation satellite scheduling problem (MOAEOSSP) often require balancing multiple conflicting objectives, primarily maximizing observation benefits and minimizing energy consumption. These objectives are inherently contradictory: increasing observation benefits generally requires more frequent attitude adjustments or additional tasks, which in turn raises energy consumption. Given limited energy resources, especially during long-term missions or under challenging conditions, managing energy consumption effectively becomes paramount. Consequently, the MOAEOSSP necessitates scheduling strategies capable of balancing multiple objectives. Achieving this balance is crucial, as it not only extends the satellite's lifespan but also enhances overall mission performance through sustainable and efficient resource allocation.

Li and Li (2019) introduced a robust model and a multi-objective binary-encoding differential evolution (MBDE) algorithm to optimize schedule profits and the slack time in agile satellite scheduling, effectively managing resource failures and emergency task insertions. Wei et al. (2021) proposed a multi-objective memetic approach (MOMA-TD) for the MOAEOSSP in a multi-orbit scenario, optimizing the failure rate and load balance with time-dependent transition times, demonstrating superior convergence and solution quality. Qiu et al. (2022) presented a framework for coupled scheduling and planning in time delay integration (TDI) imaging missions for agile satellites, optimizing the imaging resolution and energy consumption through simulations, including extreme cases. Wei et al. (2024) introduced a knowledge-transfer multi-objective genetic programming (KT-MOGP) framework to optimize profit and image quality in the dynamic MOAEOSSP, outperforming existing methods in handling real-time requests. However, the existing studies on the MOAEOSSP often overlook algorithmic parameter sensitivity, relying on fixed parameter settings that can cause instability and performance fluctuations across different scenarios. Even minor changes in these parameters can significantly impact outcomes, leading to inconsistent performance and limiting robustness and adaptability. This limitation is particularly evident in multi-objective optimization problems, which require balancing conflicting objectives and are affected by variations in the problem scale and complexity. Therefore, developing dynamic or adaptive parameter adjustment mechanisms is essential for advancing these algorithms. To address this, we use reinforcement learning to help the evolutionary algorithm dynamically adjust parameters based on changes in the problem complexity and optimization progress. This approach enables the algorithm to adapt automatically to different scenarios, enhancing robustness and adaptability in multi-objective optimization.

The independent use of reinforcement learning (RL) for satellite scheduling is still in its early stages. While a few successful applications exist, Herrmann and Schaub (2023) combined RL with Monte Carlo tree search (MCTS) for on-board AEOS planning and scheduling, improving the performance and reducing the computational time under varying target densities. Y. He et al. (2022) modeled the agile satellite scheduling problem as a Markov decision process, using a Q-network to evaluate long-term benefits. Their approach handled unknown data well and generated high total profits with strong scalability. Additionally, Ren, Ning, and Wang (2022) developed an RL-based model for thefair scheduling of agile satellite stereo imaging tasks, with experiments confirming its effectiveness in achieving optimal, fair solutions.

Although RL shows great potential in the field of satellite scheduling, it also has some significant drawbacks, such as high computational demands and an underdeveloped theoretical foundation for this application (Song et al. 2023). In contrast, meta-heuristic algorithms, such as genetic algorithm

(GA) (Çalış and Bulkan 2015), particle swarm optimization (PSO) (Sengupta, Basak, and Peters 2019), and artificial bee colony (ABC) algorithms (Karaboga, Akay, and Karaboga 2020), have a more mature theoretical basis and strong global search capabilities. These algorithms are considered intelligent because they leverage mechanisms inspired by natural evolution, social behaviors, and cooperative learning to efficiently explore complex solution spaces. Moreover, numerous studies have demonstrated their reliability and maturity in solving a wide range of optimization problems, providing a solid foundation for their application in real-world scenarios (Hsieh and Su 2016). Integrating RL into these algorithms can enhance their exploration capacity within a vast state space and enable the dynamic adjustment of search strategies. This approach of RL-assisted evolutionary algorithms has been validated across different fields, demonstrating its effectiveness in improving algorithm adaptability and optimization performance (Song, Wu, et al. 2024). For example, Hu, Gong, and Li (2021) introduced an algorithm that merges differential evolution with reinforcement learning to optimize photovoltaic (PV) model parameters, achieving better accuracy and robustness than other advanced algorithms. X. Zhang et al. (2022) proposed a multi-objective particle swarm optimization algorithm with reinforcement learning-based multi-mode collaboration (MCMOPSO-RL) for unmanned air vehicles (UAV) collaborative path planning, showing superior efficiency and robustness in complex environments. Z.-Q. Zhang et al. (2023) developed a Q-learning-based hyper-heuristic evolutionary algorithm (QHHEA) for solving distributed flexible job-shop scheduling with crane transportation (DFJSPC), outperforming state-of-the-art methods in minimizing the makespan and energy consumption. B. Zhou and Zhao (2023) presented a milk-run vehicle scheduling problem using an adaptive artificial bee colony algorithm enhanced by a deep Q-learning network (AABC-DQN), which outperforms benchmark algorithms terms of both the solution quality and convergence ability.

As outlined earlier, we discussed how existing studies have made progress in optimizing multi-objective solutions for the MOAEOSSP but still face challenges related to parameter sensitivity. To address this issue, we selected the multi-objective grey wolf optimization (MOGWO) algorithm due to its proven effectiveness and superior performance on standard benchmark problems; it has consistently outperformed other multi-objective optimization algorithms (Mirjalili et al. 2016). This strong performance, combined with its extensive application in fields such as welding production scheduling (Lu et al. 2016) and flexible job shop scheduling (Zhu and Zhou 2020), further demonstrates MOGWO's applicability and competitiveness in multi-objective optimization tasks. Its unique leader selection and archive maintenance mechanisms enhance solution diversity and convergence, making MOGWO particularly well-suited for multi-objective optimization. However, control parameters are critical for evolutionary algorithms and are often sensitive to specific problems, affecting their generalization ability across different scenarios. While RL has strong generalization capabilities, its reliance on trial-and-error exploration can lead to slower convergence and suboptimal results without explicit domain-specific knowledge, particularly in complex environments. In contrast, meta-heuristic algorithms, with their mature theoretical foundations and strong global search abilities, have already achieved good results in solving this problem. To address these issues, this study integrates RL into MOGWO, enabling adaptive parameter tuning based on the optimization progress and problem complexity. Specifically, the RL based approach assists MOGWO in dynamically adjusting its parameters to achieve a better balance between conflicting objectives, thereby enhancing robustness, adaptability, and solution quality. This novel reinforcement learning-based multi-objective grey wolf optimization algorithm not only enhances decision-making but also broadens application potential. This is also the first time RL has been applied in an evolutionary algorithm for solving the MOAEOSSP, offering new insights and methodologies for other combinatorial optimization problems.

The main contributions of this paper can be summarized as follows:

(1) We have established a novel multi-objective optimization mathematical model to address the distinct characteristics and practical engineering demands of the MOAEOSSP. This model

concurrently optimizes two pivotal objective functions: maximizing observational benefits and minimizing energy consumption.

(2) To overcome the limitations of metaheuristic algorithms, including susceptibility to local optima, parameter sensitivity, and suboptimal performance in multi-objective optimization, we propose a novel reinforcement learning-based MOGWO (RLMOGWO) algorithm. By integrating Q-learning, the algorithm balances exploration and exploitation through specific state-action combinations and dynamically adjusts parameters for better adaptability. Additionally, a reward mechanism based on the crowding distance and inverted generational distance (IGD) is introduced to maintain the diversity and distribution of Pareto front solutions, ensuring superior optimization performance.

(3) The effectiveness and applicability of RLMOGWO have been thoroughly validated through extensive simulation experiments. Compared to four multi-objective optimization algorithms, including the non-dominated sorting genetic algorithm II (NSGAII), strength Pareto evolutionary algorithm 2 (SPEA2), multi-objective evolutionary algorithm based on decomposition (MOEA/D), and standard MOGWO, the proposed algorithm demonstrated superior performance across various evaluation metrics in diverse test scenarios. These results highlight its robustness and potential for broader application in complex optimization problems.

The structure of this paper is arranged as follows: Section 2 elaborates on the detailed description and mathematical modeling of the MOAEOSSP. In Section 3, we introduce the proposed multi-objective optimization algorithm. Section 4 is dedicated to presenting the simulation experiments and providing a comprehensive analysis of the results. Finally, Section 5 encapsulates the key findings of this study and explores potential avenues for future research.

2. Mathematical model

The primary objective of satellite Earth observation scheduling is to prioritize and effectively allocate user-submitted observation requests within the constraints provided. These requests typically involve capturing images of specific regions on Earth, with task priorities established in advance by users or decision-makers. The majority of the acquired images are stored on the satellite's onboard equipment, while some are immediately transmitted to ground stations or relay satellites for processing. As a result, the AEOS must perform attitude adjustments between different tasks, with the duration of these adjustments determined by the start times and observation durations of the tasks. However, because the number of observation requests frequently exceeds the capacity of the AEOS, combined with the limitations of onboard storage space and energy resources, this issue is classified as a typical oversubscription problem.

2.1. Assumptions

Considering the high complexity of the MOAEOSSP, we address the simultaneous optimization of two primary objectives: maximizing observation benefits and minimizing energy consumption. This optimization is conducted under multiple constraints, including visibility time windows and satellite resource limitations. Given that observation requests typically exceed the AEOS capacity and are further constrained by onboard storage and energy availability, the problem is classified as an oversubscription issue. Accordingly, we have established the following reasonable assumptions to ensure the practical and effective modeling of this problem:

(1) **We focus on observable targets within a single pass.** This approach aligns with real satellite operations and simplifies scheduling by reducing complexity.

(2) **We ensure that observations are non-preemptive and exclusive.** Each task must be completed before starting a new one, reflecting practical constraints and preventing conflicts.
(3) **We ignore data downlink and recharging processes.** By excluding these processes, we simplify the model and focus on core scheduling issues.
(4) **We omit considerations of satellite malfunctions.** Assuming fault-free operation allows us to keep the model manageable and optimizable, avoiding increased complexity.

2.2. Constraints and objective function

The MOAEOSSP aims to efficiently manage multiple satellites and their respective observation tasks to optimize resource utilization and maximize task completion rates. Therefore, we define a set of symbols and variables to develop a clear model that describes the relationships between satellites and tasks:

- $Sats$: Set of satellites, $Sats = \{s_1, s_2, \ldots, s_n\}$, where n is the number of satellites.
- $Tasks$: Set of observation tasks, $Tasks = \{t_1, t_2, \ldots, t_m\}$, where m is the number of tasks.
- $Orbits_i$: Orbit set of satellite i, $Orbits_i = \{o_1, o_2, \ldots, o_q\}$, where q is the number of orbits.
- Vis_Win: Set of visible time windows, $Vis\_Win = \{v_1, v_2, \ldots\}$, where each window $v_r = \{\texttt{start}, \texttt{end}, \texttt{duration}\}$.
- Obs_Win: Set of observable time windows, $Obs\_Win = \{w_1, w_2, \ldots\}$, where each window $w_r = \{\texttt{begin}, \texttt{finish}, \texttt{length}\}$.
- N_1 represents the number of observable time windows, and N_2 represents the number of visible time windows.
- T_{jh}: Time to transfer attitude from task j to task h.
- $\texttt{Energy}$: Set of satellite energies, $Energy = \{e_1, e_2, \ldots, e_n\}$.
- $\texttt{Energy}_{ij}$: Energy consumed by satellite i for task j.
- $\texttt{E}_\texttt{i}$: Energy per unit time for attitude maneuver of satellite i.
- $\texttt{Memory}$: Set of satellite memory capacities, $\texttt{Memory} = \{m_1, m_2, \ldots, m_n\}$.
- M_{ij}: Memory used by satellite i for task j.
- $\texttt{Priority}$: Set of task priorities, $\texttt{Priority} = \{p_1, p_2, \ldots, p_m\}$.
- x_j:

$$x_j = \begin{cases} 1 & \text{if task } j \text{ is observed} \\ 0 & \text{otherwise} \end{cases}$$

- $(y_i^k)^j$:

$$(y_i^k)^j = \begin{cases} 1 & \text{if task } j \text{ is observed by satellite } i \text{ in orbit } k \\ 0 & \text{otherwise} \end{cases}$$

- $(z_i^k)_h^j$:

$$(z_i^k)_h^j = \begin{cases} 1 & \text{if task } h \text{ follows task } j \text{ on satellite } i \text{ in orbit } k \\ 0 & \text{otherwise} \end{cases}$$

Building on the above descriptions, the objective functions for the MOAEOSSP aim to maximize the observation benefit $f_1(X)$ and minimize the satellite energy consumption $f_2(X)$. Here, X is defined as the set of decision variables on which the objective functions f_1 and f_2 depend, with its specific composition determined by the established model. In Pareto optimization, the search is conducted over this set of variables to find a solution where $f_1(X)$ and $f_2(X)$ achieve Pareto optimality. These objectives are subject to constraints ensuring that each task is observed only once, the observation time falls within the satellite's visible window without overlap, and sufficient transition time is provided for attitude maneuvers, while also considering satellite resource limitations. The specific formulas are as follows:

$$\min_{X} \ \{f_1(X), f_2(X)\}, \tag{1}$$

$$f_1 = \sum_{i=1}^{n} \sum_{k=1}^{q} \sum_{j=1}^{m} p_j \cdot (y_i^k)^j, \tag{2}$$

$$f_2 = \sum_{i=1}^{n} \sum_{k=1}^{q} \sum_{j=1}^{m} \left(\texttt{Energy}_{ij} \cdot (y_i^k)^j + \sum_{h=1,h\neq j}^{m} (z_i^k)_j^h \cdot T_{jh} \cdot \mathtt{E_i} \right), \tag{3}$$

Subject to

$$\forall t_j \in \mathit{Tasks}\colon \sum_{i=1}^{n} \sum_{k=1}^{q} (y_i^k)^j \leq 1, \tag{4}$$

$$\begin{aligned} &\forall r_1, r_2 \in \{0, 1, \ldots, N_1\}, r_1 \neq r_2, l \in \{1, 2, \ldots, N_2\}, \\ &\exists\, w_{r_1,r_2} \in \mathit{Obs\_Win}, v_l \in \mathit{Vis\_Win}, \text{ suchthat:} \\ &\quad [\texttt{begin}_r, \texttt{finish}_r] \subseteq [\texttt{start}_l, \texttt{end}_l] \\ &\quad \text{and } \texttt{finish}_{r_1} \leq \texttt{begin}_{r_2}, \end{aligned} \tag{5}$$

$$\begin{aligned} &\forall t_j, t_h \in \mathit{Tasks}, j \neq h, (z_i^k)_h^j = 1: \\ &\quad \texttt{begin}_j + \texttt{length}_j + T_{jh} \leq \texttt{begin}_h, \end{aligned} \tag{6}$$

$$\begin{aligned} &\forall i \in \{1, 2, \ldots, n\}, k \in \{1, 2, \ldots, q\}, j, h \in \{1, 2, \ldots, m\}, \\ &j \neq h, \quad (y_i^k)^j = (z_i^k)_h^j = 1: \\ &\quad \sum_{k=1}^{q} \sum_{j=1}^{m} \left(\texttt{Energy}_{ij} \cdot (z_i^k)_h^j + \sum_{h=1 h\neq j}^{m} T_{jh} \cdot (z_i^k)_h^j \cdot E_i \right) \leq e_i, \\ &\quad \sum_{k=1}^{q} \sum_{j=1}^{m} M_{ij} \cdot (y_i^k)^j \leq m_i. \end{aligned} \tag{7}$$

Equation (1) represents the minimization objective function for the MOAEOSSP. Equation (2), the observation benefit function, and Equation (3), the energy consumption function, define our proposed objective functions. Constraint (4) ensures that each task is observed only once. Constraint (5) stipulates that the observable time must be within the visible window without overlapping. Constraint (6) guarantees a sufficient transition time for attitude maneuvers. Lastly, Constraint (7) represents the satellite resource constraints.

3. Methods

Many algorithms exist for solving multi-objective optimization problems, such as NSGA-II (Deb et al. 2002), SPEA2 (Zitzler, Laumanns, and Thiele 2001), and MOEA/D (Q. Zhang and Li 2007). However, these approaches often struggle with local optima, high computational demands, and parameter sensitivity. To address these challenges, we propose a novel RLMOGWO algorithm. This algorithm enhances adaptability and maintains the distribution and diversity of Pareto front solutions. By simulating the hierarchical behavior of grey wolves (Mirjalili, Mirjalili, and Lewis 2014; Mirjalili et al. 2016), RLMOGWO effectively optimizes solutions in multi-objective search spaces.

3.1. MOGWO algorithm

The social structure and hunting strategies of gray wolves serve as the core inspiration for this algorithm. In designing GWO, to accurately model the hierarchical order within a wolf pack, the best solution is designated as the α wolf. The second and third best solutions are called the β and δ wolves, respectively. In the GWO algorithm, the search process (optimization) is guided by α, β, and δ, while the ω wolves follow these three wolves to explore the global optimal solution. This hierarchical structure is illustrated in Figure 2.

According to Mirjalili et al. (2016), the position of each wolf is adjusted based on the positions of the alpha, beta, and delta wolves, guiding the overall search process. The following equations describe the update mechanism of the MOGWO algorithm in detail:

(1) The vector $\vec{D}$ is calculated to simulate the encircling behavior:

$$\vec{D} = \left|\vec{C} \cdot \vec{X}_p(t) - \vec{X}(t)\right|, \tag{8}$$

where $\vec{X}_p$ is the position of the prey, and $\vec{X}$ indicates the position vector of a grey wolf. $\vec{C}$ is the coefficient vector, and its calculation is shown in Equation (11).

(2) The position of the wolf is updated as follows:

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{D}, \tag{9}$$

Here, $\vec{A}$ is the coefficient vector, and its calculation is shown in Equation (10).

$$\vec{A} = 2\vec{a} \cdot \vec{r_1} - \vec{a}, \tag{10}$$

$$\vec{C} = 2\vec{r_2}, \tag{11}$$

where the elements of $\vec{a}$ linearly decrease from 2 to 0 over the course of iterations, and $\vec{r_1}$ and $\vec{r_2}$ are element-wise random vectors in the range [0, 1].

(3) The GWO algorithm uses social leadership and encircling behaviors to find optimal solutions. It retains the top three solutions (α, β, and δ) and forces all other agents, including omegas, to update their positions relative to these leaders. The algorithm continuously applies the following formulas to each agent during optimization:

$$\vec{D}_\alpha = \left|\vec{C}_1 \cdot \vec{X}_\alpha - \vec{X}\right|, \tag{12}$$

$$\vec{D}_\beta = \left|\vec{C}_2 \cdot \vec{X}_\beta - \vec{X}\right|, \tag{13}$$

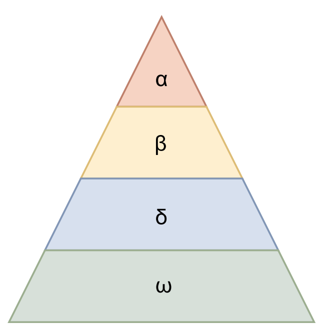


Figure 2. Grey wolf hierarchy.

$$\vec{D}_\delta = \left| \vec{C}_3 \cdot \vec{X}_\delta - \vec{X} \right|, \tag{14}$$

$$\vec{X}_1 = \vec{X}_\alpha - \vec{A}_1 \cdot \vec{D}_\alpha, \tag{15}$$

$$\vec{X}_2 = \vec{X}_\beta - \vec{A}_2 \cdot \vec{D}_\beta, \tag{16}$$

$$\vec{X}_3 = \vec{X}_\delta - \vec{A}_3 \cdot \vec{D}_\delta, \tag{17}$$

$$\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3}. \tag{18}$$

To clarify the working mechanism of MOGWO and facilitate comparisons with subsequent algorithm improvements, we present the original MOGWO pseudo-code in Algorithm 1.

Algorithm 1. Pseudo code of the MOGWO algorithm

Initialize the grey wolf population X_i ($i = 1, 2, \ldots, n$)
Initialize a, A, and C
Calculate the objective values for each search agent
Find the non-dominated solutions and initialize the archive with them
X_α = SelectLeader(archive)
Exclude alpha from the archive temporarily to avoid selecting the same leader
X_β = SelectLeader(archive)
Exclude beta from the archive temporarily to avoid selecting the same leader
X_δ = SelectLeader(archive)
Add back alpha and beta to the archive
$t = 1$
while $t <$ Max number of iterations **do**
  for each search agent **do**
    Update the position of the current search agent by equations (3.12)-(3.18)
  end for
  Update a, A, and C
  Calculate the objective values of all search agents
  Find the non-dominated solutions
  Update the archive with respect to the obtained non-dominated solutions
  if the archive is full **then**
    Run the grid mechanism to omit one of the current archive members
    Add the new solution to the archive
  end if
  if any of the newly added solutions to the archive is located outside the hypercubes **then**
    Update the grids to cover the new solution(s)
  end if
  X_α = SelectLeader(archive)
  Exclude alpha from the archive temporarily to avoid selecting the same leader
  X_β = SelectLeader(archive)
  Exclude beta from the archive temporarily to avoid selecting the same leader
  X_δ = SelectLeader(archive)
  Add back alpha and beta to the archive
  $t = t + 1$
end while
return archive

In the standard MOGWO algorithm, random initialization can lead to uneven population distribution within the search space, adversely affecting the global search capability. Furthermore, the linear decay of parameter $\vec{a}$ may not adequately balance exploration and exploitation at different stages, risking premature convergence or insufficient optimization later on, which impacts the algorithm's stability and precision. To address these limitations, we propose several improvements to enhance the performance of the MOGWO algorithm.

(1) During the algorithm initialization process, the piecewise linear chaotic map (PWLCM) (H. Zhou and Ling 1997) is used to generate the initial population, ensuring a uniform distribution of individuals within the search space and thereby enhancing population diversity. The PWLCM is defined as follows (Equation 19):

$$x(t+1) = \begin{cases} \frac{x(t)}{p_1}, & 0 \leq x(t) < p_1, \\ \frac{x(t)-p_1}{p_2-p_1}, & p_1 \leq x(t) < p_2, \\ \frac{1-x(t)}{1-p_2}, & p_2 \leq x(t) < 1. \end{cases} \tag{19}$$

where $x(t)$ is the state variable of the Piecewise Linear Chaotic Map (PWLCM), defined within the range [0, 1) and updated iteratively based on piecewise linear rules. The parameters p_1 and p_2 are the control parameters of the map, both belonging to the interval $0 \leq p \leq 1$. In this case, p_1 is set to 0.3 and p_2 is set to 0.7.

This method broadens the diversity of the initial population, laying a solid foundation for optimization. By examining the Euclidean distance from each point to the centroid, we gain a deeper understanding of the distribution of initial solutions within the search space, which is crucial for effective exploration. As shown in Figure 3 (the red bars and lines represent PWLCM, while the blue bars and lines represent random initialization), with a population size of 50, each initialization method was run 30 times with 100 iterations each to obtain more stable and representative distribution characteristics. Multiple runs reduce the influence of randomness, making the results more reliable. In terms of the distribution, PWLCM initialization exhibits a more uniform and extensive spread, covering a wider range within the search space, indicating higher population diversity. In contrast, random initialization shows a more concentrated distribution with relatively narrow coverage. Regarding peaks and spread, PWLCM initialization has a flatter and broader peak, which helps prevent premature convergence and enables a more comprehensive search of the space, whereas random initialization has concentrated peaks, potentially leading to premature convergence. The density curves further validate this difference: PWLCM's density curve is flatter and more dispersed, facilitating thorough search space coverage, while random initialization's curve is concentrated and quickly drops off, indicating a higher risk of becoming trapped in local optima. Consequently, PWLCM's uniformity and dispersion enhance the optimization process's capacity for global solution exploration.

Subsequently, opposition-based learning (Tizhoosh 2005) is applied and combined with the original population. Through non-dominated sorting, the top N individuals are selected as the

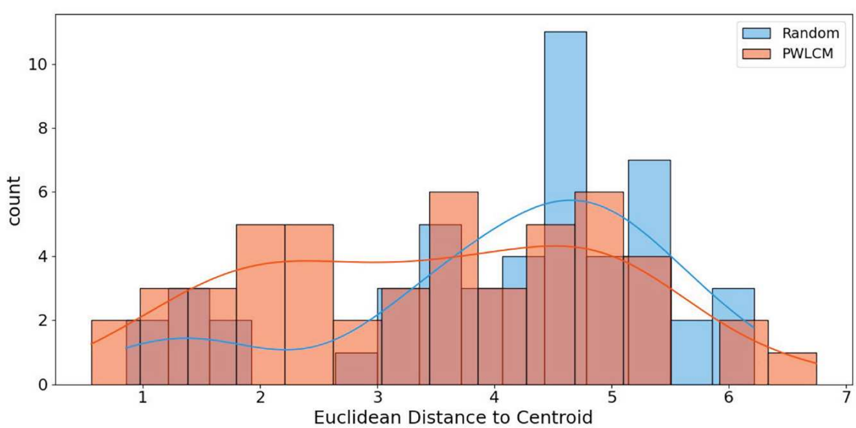


Figure 3. Euclidean distance distribution for Random vs. PWLCM initialization.

new initial population. Opposition-based learning is defined as follows (Equation 20):

$$X_{obl} = k \cdot (U_b + L_b) - X, \tag{20}$$

where k is a random variable drawn from a normal distribution, U_b and L_b represent the upper and lower bounds of the search space, and X is the current position of the solution.

(2) Replacing the linear decay of $\vec{a}$ in the original literature with a cosine-based variation that reflects natural periodic patterns, as shown in Equation (21), can more effectively balance exploration and exploitation. Here, t is the current iteration, and T is the maximum number of iterations:

$$\vec{a} = 2 \times \cos\left(\frac{\pi t}{2T}\right). \tag{21}$$

This approach strengthens exploration in the early stages and gradually enhances exploitation later, reducing the likelihood of getting stuck in local optima. The smooth transition provided by the cosine function also improves global search ability, robustness, and adaptability, leading to faster convergence and higher accuracy across different problems.

3.2. RL-based MOGWO algorithm

The MOGWO algorithm is highly sensitive to the parameter configuration, making its performance reliant on specific problems and scenarios. Reconfiguring parameters can be time-consuming when conditions change. To address this, we incorporate Q-learning for adaptive parameter optimization, enhancing the algorithm's adaptability. Q-learning, a reinforcement learning technique with a solid theoretical foundation and straightforward implementation, has been successfully applied in models like deep Q networks (DQNs) and deep reinforcement learning (DRL). By continuously interacting with the environment, Q-learning updates the Q-values of state-action pairs, enabling optimal action selection. This adaptive mechanism not only helps the algorithm learn effective strategies in complex environments but also increases its flexibility and robustness in dynamic scenarios, ultimately improving practical effectiveness (Clifton and Laber 2020; Kumar et al. 2020; Sutton and Barto. 2018).

Q-learning involves five key components: actions, states, rewards, RL-policy, and the environment, as shown in Figure 4. This method updates Q-values to determine population update strategies, effectively replacing the random variations of A and C in MOGWO. The agent selects actions

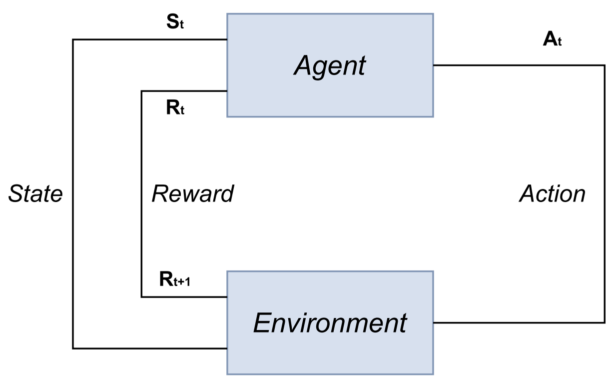


Figure 4. Five elements of Q-learning.

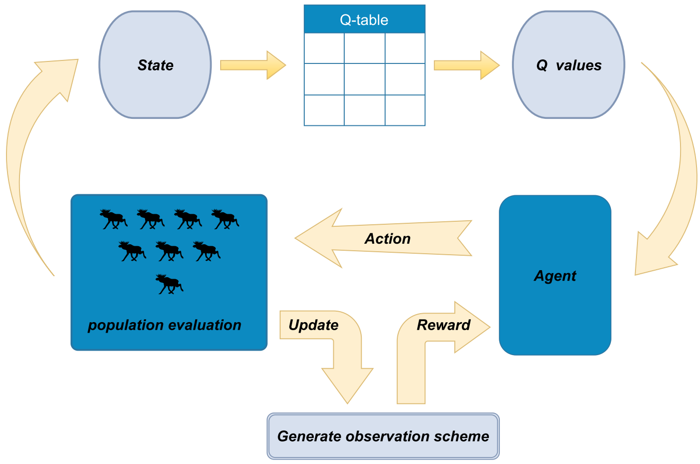


Figure 5. Q-learning method decision-making process for MOAEOSSP.

based on Q-values and action selection strategies, which then adjust the values of A and C. The reward function provides feedback based on changes in the CD and The IGD (Sun, Yen, and Yi 2019), as defined in Equations (22) and (23), to evaluate the interaction between the agent and the environment. The CD and IGD assess the diversity and convergence of the solution set, respectively. A larger CD indicates a more evenly distributed solution set, while a smaller IGD signifies that the solution set is closer to the ideal Pareto front. A decrease in the CD may lead to reduced diversity, whereas a decrease in the IGD typically indicates improved optimization. Therefore, balancing these two metrics in multi-objective optimization is essential to ensure the solution set remains well-distributed and close to the optimal solution. The magnitude of the reward directly influences the balance between exploration and exploitation. Figure 5 illustrates the Q-learning process, where continuous Q-value updates enable the agent to learn the optimal population update strategy, thereby improving the overall performance of the optimization algorithm:

$$CD = \frac{1}{N-2} \sum_{n=2}^{N-1} \sum_{m=1}^{M} \left(\frac{f_{n+1}^m - f_{n-1}^m}{f_{\max}^m - f_{\min}^m} \right), \tag{22}$$

- N is the total number of solutions.
- M is the number of objective functions.
- f_{n+1}^m and f_{n-1}^m are the objective function values of the solutions adjacent to solution n after sorting for the mth objective function.
- $f_{\max}^m$ and $f_{\min}^m$ are the maximum and minimum values of the mth objective function.
- N−2 represents the number of non-boundary solutions, used to calculate the average crowding distance for these solutions.

$$IGD = \frac{1}{|P^*|} \sum_{p \in P^*} \min_{q \in Q} \|p - q\|. \tag{23}$$

- P^* is the ideal Pareto front.
- Q is the current set of solutions generated by the algorithm.
- $\|p - q\|$ is the Euclidean distance between the solution p and the solution q.
- $|P^*|$ is the number of points in the ideal Pareto front.

As shown in Figure 5, the RLMOGWO algorithm incorporates Q-learning to optimize the agent's search strategy through a continuous cycle of interactions. In this framework, we assume that the environment satisfies the Markov property, meaning that each state transition depends solely on the current state and the action taken, without relying on the history of previous states. Specifically, the state space S is defined based on changes in the CD and IGD between consecutive iterations, capturing all essential information needed for the multi-objective optimization process. The agent perceives the current environmental state and selects the optimal action based on the Q-values in the Q-table, which adjusts parameters $\vec{r_1}$ and $\vec{r_2}$ to directly influence the values of the CD and IGD in the next state. The reward function provides feedback based on changes in the CD and IGD to balance exploration and exploitation. An increase in the CD reflects a more evenly distributed solution set, while a decrease in the IGD suggests that the solution set is closer to the ideal Pareto front. Balancing these two metrics is essential in multi-objective optimization, ensuring that the solution set remains both well-distributed and close to the optimal solution. By continuously updating Q-values, the agent learns an optimal population update strategy under a Markov decision process (MDP), allowing RLMOGWO to dynamically adjust search strategies in changing environments and achieve more efficient multi-objective optimization.

The updated formula for the Q value is presented below:

$$Q_{t+1}(S_t, A_t) = Q_t(S_t, A_t) + \alpha\left(R_t + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t)\right), \tag{24}$$

$$A_t = \begin{cases} \text{random action} & \text{if rand} < \epsilon \\ \arg\max_a Q_t(S_t, a) & \text{otherwise.} \end{cases} \tag{25}$$

where S_t is the state at time step t, A_t is the action taken at time t, R_t is the reward received after taking action A_t, α is the learning rate, γ is the discount factor, and ε is the exploration rate.

Based on the characteristics of MOGWO, we have designed a system incorporating states, actions, and a reward mechanism. States represent different conditions of the system, actions are the executable operations, and the reward mechanism provides feedback based on the outcomes of these actions, helping to optimize the algorithm:

(1) State Space S: The state space S describes the search agent's condition at each iteration t. The states are defined based on the changes in ΔCD and ΔIGD between consecutive iterations. Specifically, t represents the current iteration, and the states are categorized into four types θ as follows:

$$\begin{aligned} \Delta CD &= CD(t) - CD(t-1), \\ \Delta IGD &= IGD(t-1) - IGD(t), \end{aligned} \tag{26}$$

$$\theta = \begin{cases} \theta_1, & \text{if } \Delta CD > 0 \text{ and } \Delta IGD > 0 \\ \theta_2, & \text{if } \Delta CD > 0 \text{ and } \Delta IGD \le 0 \\ \theta_3, & \text{if } \Delta CD \le 0 \text{ and } \Delta IGD > 0 \\ \theta_4, & \text{if } \Delta CD \le 0 \text{ and } \Delta IGD \le 0, \end{cases} \tag{27}$$

(2) Reward Function R: The reward function R assigns rewards based on the state θ. The reward values are set to guide the algorithm toward more optimal solutions, where the highest rewards are given when the diversity and quality of the solutions improve:

$$R(\theta) = \begin{cases} 10, & \text{if } \theta_1 \\ 5, & \text{if } \theta_2 \\ 3, & \text{if } \theta_3 \\ 1, & \text{if } \theta_4, \end{cases} \tag{28}$$

(3) Action Space A: The action space A includes the different actions that can be taken in each state. These actions adjust the parameters r_1 and r_2, which in turn influence the values of $\vec{A}$ and $\vec{C}$, altering the balance between exploration and exploitation. The action space is defined as follows:

$$A = \{a_1, a_2, \ldots, a_{10}\}. \tag{29}$$

Each action corresponds to different adjustments for r_1 and r_2:

$$\begin{aligned}
&a_1{:}r_1, r_2 \in [0.7, 1.0],\\
&a_2{:}r_1, r_2 \in [0.0, 0.3],\\
&a_3{:}r_1, r_2 \in [0.3, 1.0],\\
&a_4{:}r_1, r_2 \in [0.0, 0.5],\\
&a_5{:}r_1, r_2 \in [0.5, 1.0],\\
&a_6{:}r_1, r_2 \in [0.0, 1.0],\\
&a_7{:}r_1, r_2 \in [0.9, 1.0],\\
&a_8{:}r_1, r_2 \in [0.0, 0.1],\\
&a_9{:}r_1, r_2 \in [0.6, 1.0],\\
&a_{10}{:}r_1, r_2 \in [0.0, 0.4].
\end{aligned}$$

The 10 actions can be categorized into three types: exploration-enhancing actions (a_1 and a_7), which increase r_1 and r_2 to boost search randomness and expand the exploration of the solution space; balanced exploration and exploitation (a_3, a_5,a_6 and a_9), which adjust r_1 and r_2 in a moderate range to balance exploration and exploitation, suitable for the transition phase of the search; and exploitation-enhancing actions (a_2, a_4, a_8 and a_{10}), which decrease r_1 and r_2 to reduce search randomness, focusing on local exploitation and refining the search towards convergence to the optimal solution.

Building on Algorithm 1, we have incorporated the aforementioned improvement strategies and present the pseudo-code of our proposed RLMOGWO algorithm in Algorithm 2.

Algorithm 2. RLMOGWO

Initialize the grey wolf population X_i ($i = 1, 2, \ldots, n$)
Initialize a, A, C, and Q-table
Set learning rate α, discount factor γ, initial exploration rate ε
Set minimum exploration rate ϵ_{min} and exploration decay rate ϵ_{decay}
Calculate the objective values for each search agent
Find the non-dominated solutions and initialize the archive with them
$X_\alpha \leftarrow$ SelectLeader(archive)
Exclude α from the archive temporarily to avoid selecting the same leader
$X_\beta \leftarrow$ SelectLeader(archive)
Exclude β from the archive temporarily to avoid selecting the same leader
$X_\delta \leftarrow$ SelectLeader(archive)
Add back α and β to the archive
$t \leftarrow 1$
while $t <$ Max number of iterations **do**
  for each search agent **do**
    Observe current state θ_t based on ΔCD and ΔIGD
    if random number $< \epsilon$ **then**
      Select a random action A_t from the action space
    else
      Select the action A_t with the highest Q-value from the Q-table
    end if
    Adjust r_1 and r_2 according to action A_t
    Update the position of the current search agent by Equations (12)–(18)

end for
Update a, A, and C
Calculate the objective values of all search agents
Find the non-dominated solutions
Update the archive with respect to the obtained non-dominated solutions
if the archive is full **then**
    Run the grid mechanism to omit one of the current archive members
    Add the new solution to the archive
end if
if any of the newly added solutions to the archive is located outside the hypercubes **then**
    Update the grids to cover the new solution(s)
end if
$X_\alpha \leftarrow$ SelectLeader(archive)
Exclude a from the archive temporarily to avoid selecting the same leader
$X_\beta \leftarrow$ SelectLeader(archive)
Exclude β from the archive temporarily to avoid selecting the same leader
$X_\delta \leftarrow$ SelectLeader(archive)
Add back a and β to the archive
for each search agent **do**
Calculate reward R_t based on state θ_t
Update Q-value in Q-table using:
$Q(\theta_t, A_t) \leftarrow Q(\theta_t, A_t) + \alpha \times (R_t + \gamma \times \max_a Q(\theta_{t+1}, a) - Q(\theta_t, A_t))$
end for
Decay epsilon: $\epsilon \leftarrow \max(\epsilon_{min}, \epsilon \times \epsilon_{decay})$
$t \leftarrow t + 1$
end while
return archive

4. Results

4.1. Experimental settings

Due to the lack of standard test cases for optimizing the MOAEOSSP and the difficulty of obtaining real-world satellite scheduling data, we generated a set of reliable test data using simulation software. The scheduling period is set to May 31, 2024, 04:00:00 to 07:00:00, assuming that satellites are operated by Chinese authorities. The observation area covers the Chinese mainland, with a latitude range of $30^\circ N \leq \text{latitude} \leq 40^\circ N$ and a longitude range of $90^\circ E \leq \text{longitude} \leq 110^\circ E$. In this study, we utilized a constellation of 10 satellites, where each satellite's orbit is defined by six key orbital elements: the inclination (i), right ascension of the ascending node, eccentricity, argument of perigee, initial true anomaly, and orbit radius. The initial orbital parameters for each satellite are set as follows: an inclination of 30°, a right ascension of the ascending node of 0°, an eccentricity of 0, an argument of perigee of 0, an initial true anomaly of 60°, and an orbit radius of 7200 km. Each satellite's total energy is set to 200, with an energy consumption rate of 3 units per unit time for attitude maneuvers, simulating realistic energy constraints during task scheduling. We created multiple scenarios containing 50 to 300 tasks, with increments of 50 tasks. The scenario are called `MO_case_50`, `MO_case_100`, `MO_case_150`, `MO_case_200`, `MO_case_250`, and `MO_case_300`, respectively. The simulation software calculated all feasible time windows for satellite visibility to ground stations and processed the resulting data to generate the available time windows for each task. Additionally, task priorities were randomly assigned within the range of [1, 10]. The simulation experiments were conducted in a hardware environment with an Intel(R) Core(TM) i5-1340P processor (base frequency 1.9GHz, 16 cores) and 16 GB RAM, running on a 64-bit Windows 11 system, and the programming was done using MATLAB 2023b.

4.2. Parameter settings

To validate the effectiveness of the proposed algorithm, we selected NSGAII (Deb et al. 2002), SPEA2 (Zitzler, Laumanns, and Thiele 2001), MOEA/D (Q. Zhang and Li 2007), and

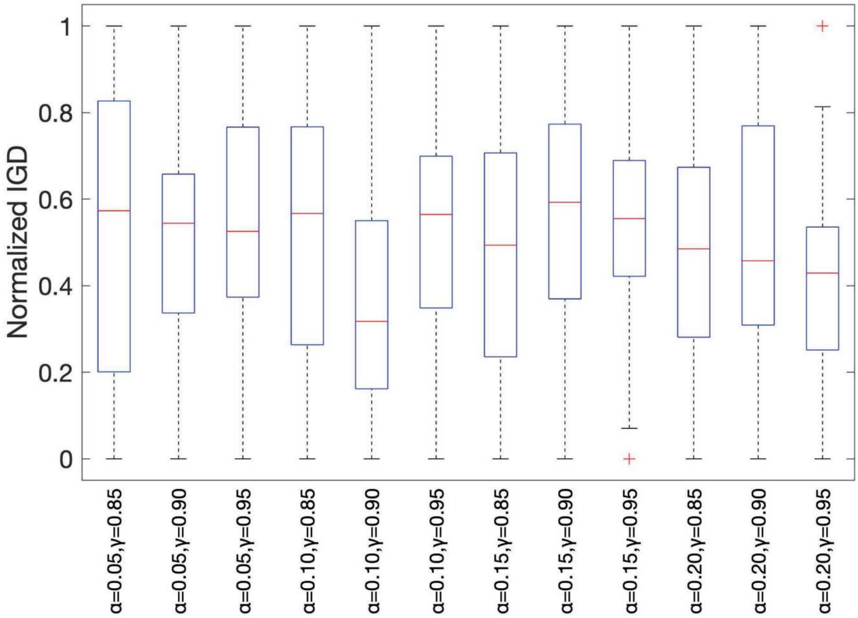


Figure 6. Boxplot of normalized IGD for different hyperparameter settings (α and γ) over 30 runs.

MOGWO (Mirjalili et al. 2016) for comparison.The performance of the proposed algorithm was comprehensively evaluated based on these algorithms, and the comparison parameter settings were based on the recommendations in the above references. The population size was 50, the external archive size was 150, the number of grids was 30, and the maximum number of iterations was 100.

For Q-learning, the initial epsilon for the epsilon-greedy strategy was set to $\epsilon = 1.0$, with a minimum epsilon of $\epsilon_{\min} = 0.01$ and an epsilon decay rate of $\epsilon_{\text{decay}} = 0.995$. The learning rate ($\alpha$) and discount factor ($\gamma$) were set to 0.1 and 0.9, respectively. These hyperparameters were initially selected based on commonly used values in reinforcement learning and by drawing on insights from reference (Song et al. 2023). To simplify the experimental setup, we performed a reasonable simplification of the hyperparameter selection process, focusing on configurations that provide a balance between convergence and stability. To further evaluate the impact of different hyperparameter combinations, we conducted a sensitivity analysis, as shown in Figure 6. The boxplot summarizes the Normalized IGD values across 30 independent runs under the *MO_case_150* scenario. Results demonstrate that the combination $\alpha = 0.1$ and $\gamma = 0.9$ yields relatively stable and low IGD values. However, variations in other parameter settings indicate the sensitivity of Q-learning to hyperparameter choices. It is worth noting that these findings are scenario-specific, and further exploration, including systematic tuning and multi-scenario validation, will be conducted in future research to enhance the algorithm's robustness and generalizability.

4.3. Multi-objective evaluation metrics

Three widely used metrics for evaluating algorithm performance are the IGD (Sun, Yen, and Yi 2019), hypervolume (HV) (Chen et al. 2020), and Pareto front (PF) approximation error (Δ_p) (Schutze et al. 2012). Their definitions are shown in Equations (23), (30) and (31).

The IGD measures the average distance from each point on the ideal PF P^* to the nearest solution in the current set Q, with smaller values indicating better convergence and distribution.

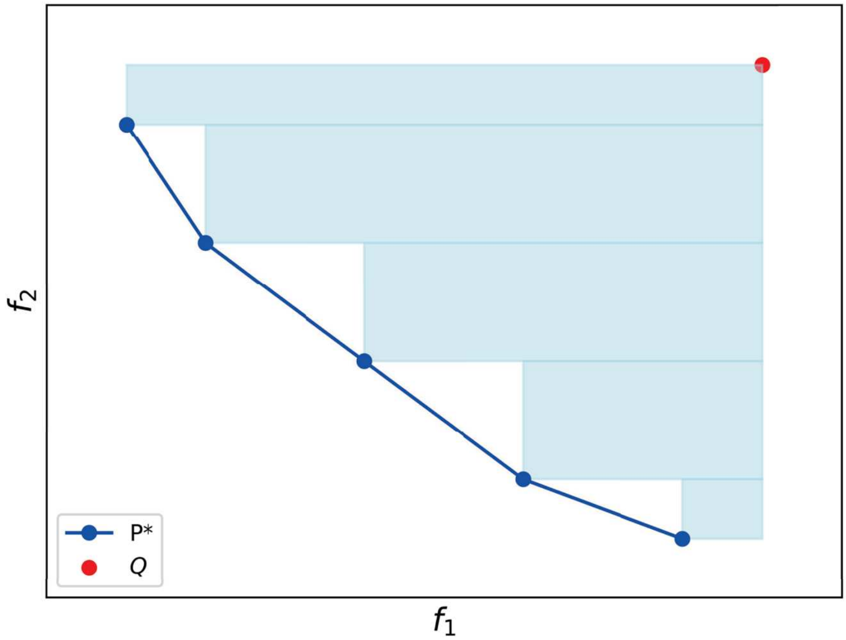


Figure 7. Schematic diagram of HV index calculation of dual objective function.

The HV measures the hypervolume of the objective space formed by the ideal Pareto front P^* and the obtained solutions Q, as shown by the blue area in Figure 7. Larger values indicate better convergence and diversity.

$$\mathrm{HV} = \mathrm{Volume}\left(\bigcup_{q\in Q} \mathrm{dominated}(q, P^*)\right), \tag{30}$$

- P^* is the ideal PF, serving as the reference point in the objective space.
- Q is the current set of solutions generated by the algorithm.
- dominated(q, P^*) is the region in the objective space that is dominated by the solution q with respect to the ideal Pareto front P^*.
- Volume is the volume operator, calculating the hypervolume of the union of all dominated regions.

Δ_p, combining the modified generational distance (GD) and IGD, measures the average Hausdorff distance between the obtained solutions Q and the ideal Pareto front P^*; smaller Δ_p values indicate better algorithm performance.

$$\Delta p = \frac{1}{|Q|-1}\sum_{i=1}^{|Q|-1}\left|\|q_i - q_{i+1}\| - \frac{1}{|Q|-1}\sum_{j=1}^{|Q|-1}\|q_j - q_{j+1}\|\right|. \tag{31}$$

- $|Q|$ is the number of solutions in the current set Q.
- q_i is the ith solution in the sorted solution set Q.
- $\|q_i - q_{i+1}\|$ is the Euclidean distance between consecutive solutions q_i and q_{i+1} in the solution set Q.

- $\frac{1}{|Q|-1}\sum_{j=1}^{|Q|-1}\|q_j - q_{j+1}\|$ is the average Euclidean distance between consecutive solutions in Q.
- $\left|\|q_i - q_{i+1}\| - \frac{1}{|Q|-1}\sum_{j=1}^{|Q|-1}\|q_j - q_{j+1}\|\right|$ is the absolute difference between the distance of two consecutive solutions and the average distance in Q.

4.4. Simulation results

To evaluate the effectiveness of the Q-learning policy, the Reward vs Iteration plot (Figure 8) is presented to illustrate the learning process and convergence behavior. As shown in the plot, the Q-learning policy undergoes a dynamic learning process. In the initial stages, the reward exhibits noticeable fluctuations, reflecting the exploration phase where the algorithm actively searches the solution space and adjusts its policy. However, as the learning progresses, these fluctuations gradually diminish, signaling a transition from exploration to exploitation. Eventually, the reward stabilizes, indicating that the Q-learning policy has successfully converged to an optimal or near-optimal solution.

Figure 9 illustrates the non-dominated solution sets (NDSs) across six different scenarios, where the x-axis represents the objective function f_1 (maximizing the observation benefit) and the y-axis represents the objective function f_2 (minimizing the energy consumption), both of which are normalized to 1. In these various scenarios, the proposed **RLMOGWO** algorithm outperforms all other algorithms in terms of solution quality, demonstrating its ability to enhance the observation benefit while effectively reducing the energy consumption.

To ensure the reliability of the experimental results, each algorithm was independently executed 30 times for every scenario. The values of two objective functions were recorded, along with the evaluation outcomes for three metrics: the IGD, the HV, and Δ_p. The mean and standard deviation were calculated for each. Since the ideal PF is unknown for each test scenario, we aggregated all the NDSs from the 30 runs of each algorithm to construct an approximate ideal PF for each scenario.

The results of the objective function evaluations are presented in Table 1. Table 2 summarizes the IGD values for each algorithm across different scenarios, while Table 3 displays the corresponding HV values. Finally, the Δ_p metric results are detailed in Table 4. These tables provide a

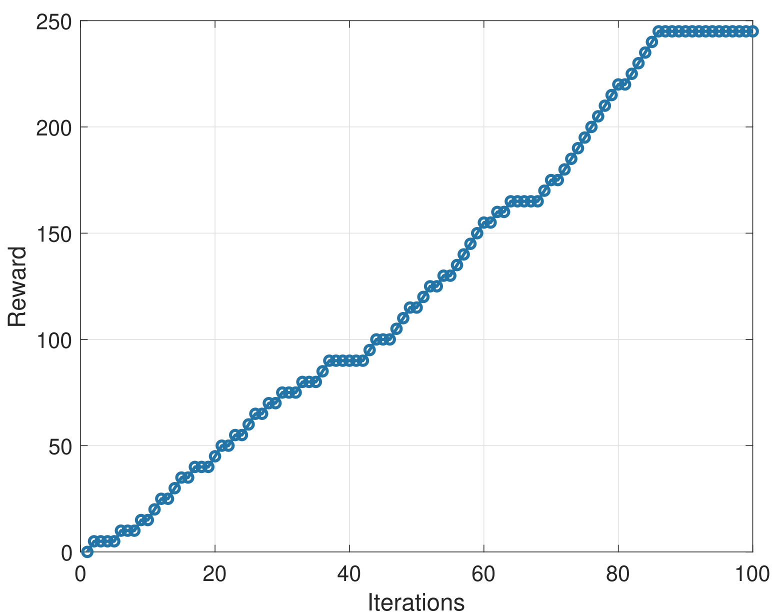


Figure 8. Variation curve of reward vs iteration.

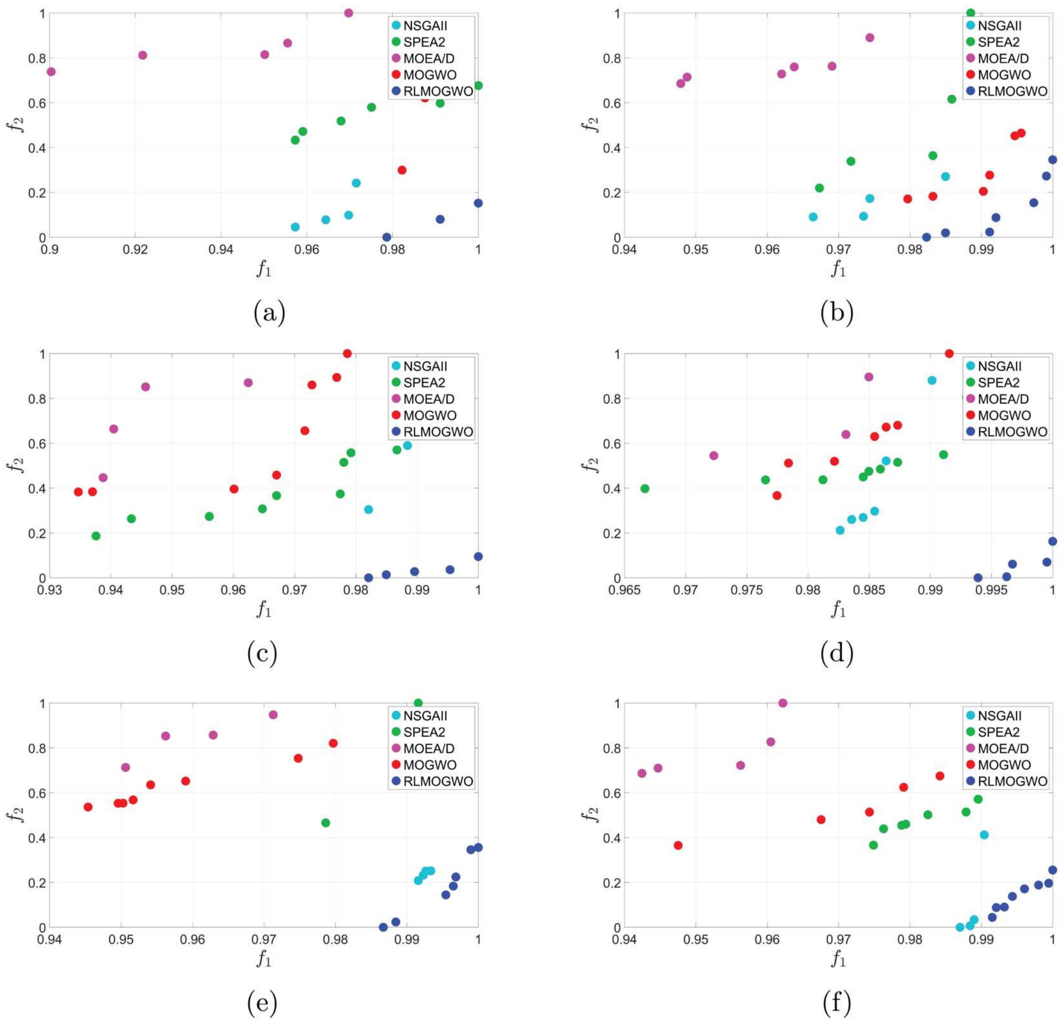


Figure 9. RLMOGWO and comparison algorithms retain NDSs in different MOAEOSSP scenarios. (a) NDSs for `MO_case_50`. (b) NDSs for `MO_case_100`. (c) NDSs for `MO_case_150`. (d) NDSs for `MO_case_200`. (e) NDSs for `MO_case_250` and (f) NDSs for `MO_case_300`.

comprehensive comparison of the performances of the algorithms across all scenarios, highlighting their strengths and weaknesses in different aspects of multi-objective optimization.

According to the data in Table 1, **RLMOGWO** stands out as the most effective algorithm, particularly in optimizing the objective function f_1. It consistently achieves or approaches ideal values across all scenarios with low variance, indicating highly stable performance. Moreover, RLMOGWO also demonstrates strong capabilities in minimizing f_2 in several scenarios, making it a robust choice for comprehensive optimization tasks. **NSGAII** exhibits superior performance in minimizing f_2, achieving the lowest average values in multiple scenarios, although its stability is comparatively lower. **SPEA2**, on the other hand, is distinguished by its high stability in f_2 optimization, making it valuable for tasks where consistency in f_2 results is critical, despite its less competitive f_1 optimization performance compared to RLMOGWO. In contrast, **MOEA/D** and **MOGWO** are generally outperformed by other algorithms; however, they show some stability in f_2 optimization under specific scenarios. Overall, RLMOGWO is well-suited for a wide range of optimization tasks, particularly those that emphasize f_1 optimization and result stability, while

Table 1. Algorithm comparison: average and standard deviation of objectives.

Scenario	Objective	Algorithm									
		NSGAII		SPEA2		MOEA/D		MOGWO		RLMOGWO	
		Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std
	f1	0.6897	0.1237	0.8103	0.1169	0.3244	0.1557	0.7192	0.1522	**1.0000**	**0.0000**
	f2	**0.3214**	0.1805	0.5174	**0.0795**	0.8551	0.0862	0.4830	0.1897	0.3338	0.1344
	f1	0.6505	0.1410	0.7283	0.0848	0.3364	0.1739	0.6242	0.1371	**0.9586**	**0.0731**
	f2	**0.2313**	0.1517	0.5582	0.0788	0.8951	**0.0701**	0.5616	0.1406	0.3125	0.1168
	f1	0.6430	0.1324	0.7883	0.0439	0.2765	0.1307	0.5780	0.1411	**0.9969**	**0.0131**
	f2	0.3313	0.1423	0.5383	**0.0636**	0.8450	0.0675	0.5982	0.0660	**0.3179**	0.1195
	f1	0.6457	0.1201	0.6760	0.0908	0.2403	0.1421	0.5558	0.0773	**0.9597**	**0.0149**
	f2	**0.3807**	0.1580	0.6094	0.0694	0.9023	0.0722	0.6268	**0.0460**	0.3839	0.0944
	f1	0.7311	0.2097	0.7757	0.0658	0.2330	0.1217	0.7013	0.0977	**1.0000**	**0.0000**
	f2	0.3328	0.1445	0.5698	0.0859	0.8548	0.0848	0.6004	**0.0693**	**0.2471**	0.1041
	f1	0.5733	0.1155	0.7402	0.0687	0.1703	0.1033	0.5490	0.1376	**0.9230**	**0.0648**
	f2	0.4833	0.1644	0.6537	**0.0407**	0.9045	0.0608	0.6734	0.0596	**0.4595**	0.1176

Table 2. Comparison of IGD values of the five algorithms.

Scenario	Algorithm									
	NSGAII		SPEA2		MOEA/D		MOGWO		RLMOGWO	
	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std
	0.1223	0.0953	0.1353	0.0701	0.6220	0.2114	0.2131	0.1281	**0.0502**	**0.0527**
	0.1688	0.1421	0.1851	0.0897	0.5176	0.2073	0.2592	0.1422	**0.0267**	**0.0225**
	0.1639	0.1286	0.1263	0.0767	0.6659	0.2403	0.2726	0.1465	**0.0323**	**0.0312**
	0.1541	0.1130	0.1839	0.0739	0.5741	0.2151	0.2725	0.1106	**0.0558**	**0.0672**
	0.0924	0.0680	0.1372	0.0798	0.6424	0.2067	0.1810	0.0774	**0.0402**	**0.0273**
	0.1021	0.0836	0.1963	0.1021	0.6500	0.2279	0.3630	0.1995	**0.0703**	**0.0796**

This table summarizes the mean and standard deviation of the IGD for different algorithms across various scenarios, both normalized to 1.

NSGAII and SPEA2 also present competitive advantages in f_2 optimization, depending on the stability requirements of the task.

As indicated by Table 2, it is evident that **RLMOGWO** consistently demonstrated superior performance in minimizing the IGD values across all scenarios, largely due to the integration of the the IGD into the Q-learning state space. This integration allowed the algorithm to more effectively approximate the ideal Pareto front, leading to significantly lower IGD values compared to other algorithms. By incorporating the IGD into the decision-making process, RLMOGWO was able to navigate the solution space more efficiently, resulting in closer proximity to the optimal solutions and greater stability, particularly in more complex scenarios. This strategic enhancement of RLMOGWO made it the most effective algorithm in achieving minimal IGD, outperforming others like **MOEA/D**, which struggled to maintain similar proximity to the ideal front.

As shown in Table 3, **RLMOGWO** achieved the highest HV values in most scenarios (from `MO_case_150` to `MO_case_300`), particularly excelling as the complexity of the scenarios increased. Not only did RLMOGWO's HV values surpass those of other algorithms, but its standard deviations were also typically the lowest, indicating a high level of stability while maintaining broad coverage of the objective space. **NSGAII** performed well in the first two simpler scenarios (`MO_case_500` and `MO_case_100`), achieving the highest HV values; however, as the scenario complexity increased, its performance was gradually surpassed by that of RLMOGWO. This demonstrates that RLMOGWO holds significant advantages in term of both solution quality and stability, especially when dealing with more complex multi-objective optimization problems.

From Table 4, it is evident that **RLMOGWO** consistently achieves the best Δ_p values across all scenarios, highlighting its exceptional ability to ensure both the uniformity and diversity of solution distributions. The optimal Δ_p values indicate that RLMOGWO can generate a solution set that is evenly distributed in the objective space while covering a broader range of the Pareto front. Furthermore, although **SPEA2** demonstrates higher stability in the `MO_case_300` scenario, RLMOGWO still maintains a lower average Δ_p value, further confirming its ability to achieve uniformly distributed solutions while consistently delivering superior performance across various scenarios.

Based on the analysis of the objective functions and theIGD, HV, and Δ_p metrics, **RLMOGWO** consistently demonstrated superior performance across a wide range of scenarios. It excelled in maximizing the objective function values and minimizing the IGD, indicating its ability to generate solutions that closely approximate the ideal Pareto front. Additionally, RLMOGWO achieved the highest HV values in most scenarios, particularly as the complexity increased, showcasing its ability to maintain broad coverage of the objective space while ensuring stability. Furthermore, its consistently low Δ_p values highlight RLMOGWO's proficiency in producing uniformly distributed solutions with extensive diversity. While **NSGAII** and **SPEA2** showed strengths in specific metrics and simpler scenarios, RLMOGWO outperformed them as the problem complexity increased, making it the most effective and dependable algorithm for multi-objective optimization tasks across different scenarios.

Table 3. Comparison of HV values of the five algorithms.

Scenario	Algorithm									
	NSGAII		SPEA2		MOEA/D		MOGWO		RLMOGWO	
	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std
	9.0964e−02	1.6889e−04	9.0658e−02	1.3914e−04	9.0479e−02	1.3003e−04	9.0691e−02	1.6944e−04	9.0942e−02	**5.2490e−05**
	9.1008e−02	1.0167e−04	9.0761e−02	8.4471e−05	9.0673e−02	1.0680e−04	9.0733e−02	1.2466e−04	9.0993e−02	**5.1256e−05**
	9.0983e−02	5.8916e−05	9.0834e−02	8.4946e−05	9.0766e−02	6.5107e−05	9.0828e−02	1.0169e−04	**9.0987e−02**	**4.8948e−05**
	9.0961e−02	8.4018e−05	9.0839e−02	5.0950e−05	9.0743e−02	6.6048e−05	9.0799e−02	5.1467e−05	**9.0968e−02**	**3.8762e−05**
	9.0912e−02	7.7125e−05	9.0836e−02	6.6670e−05	9.0773e−02	7.3988e−05	9.0807e−02	7.8732e−05	**9.0958e−02**	**4.0352e−05**
	9.0952e−02	7.1995e−05	9.0887e−02	6.2576e−05	9.0879e−02	9.3139e−05	9.0857e−02	9.2633e−05	**9.0974e−02**	**4.4272e−05**

This table summarizes the mean and standard deviation of the HV for different algorithms across various scenarios.

Table 4. Comparison of Δ_p values of the five algorithms.

Scenario	Algorithm									
	NSGAII		SPEA2		MOEA/D		MOGWO		RLMOGWO	
	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std
	0.1346	0.1060	0.2233	0.1353	0.5899	0.1621	0.1998	0.1467	**0.0456**	**0.0456**
	0.1512	0.1225	0.2639	0.1430	0.5968	0.1667	0.2895	0.1315	**0.0351**	**0.0425**
	0.1384	0.0951	0.1588	0.0753	0.5470	0.1861	0.2726	0.1641	**0.0346**	**0.0251**
	0.1508	0.1242	0.2647	0.1227	0.6210	0.1815	0.3624	0.1308	**0.0716**	**0.0766**
	0.0832	0.0639	0.2457	0.2044	0.5101	0.1699	0.2943	0.1709	**0.0302**	**0.0250**
	0.1462	0.1342	0.1127	**0.0611**	0.6231	0.2356	0.2164	0.1413	**0.0593**	0.0778

This table summarizes the mean and standard deviation of Δ_p for different algorithms across various scenarios, both normalized to 1.

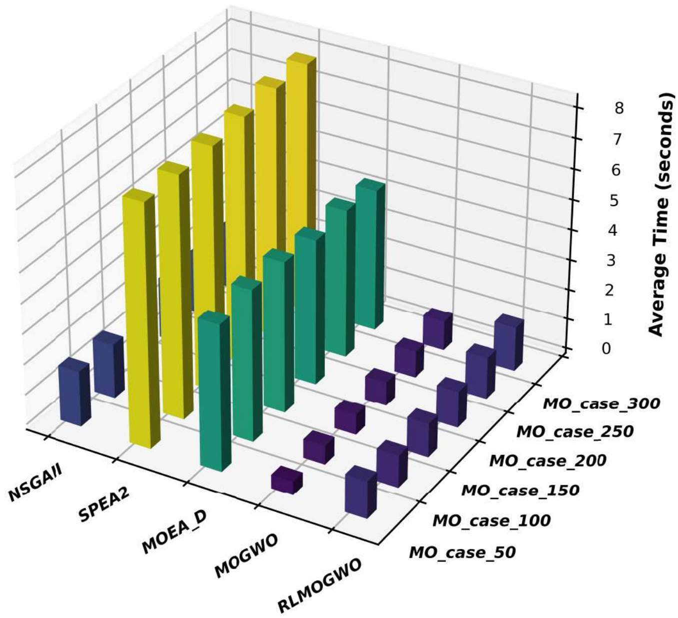
In six different scenarios, we conducted 30 runs for each of the five algorithms, evaluating their execution time and memory usage to analyze the efficiency and stability of each algorithm. This test aimed to provide a comprehensive assessment of the algorithms' performance, with a particular focus on time efficiency and resource utilization. Figure 10 presents a 3D bar chart comparing the execution time and memory usage of each algorithm across all scenarios, providing an overall comparison of their performance. To gain a deeper understanding of each algorithm's behavior in specific contexts, we conducted a detailed analysis of a medium-scale scenario. Figure 11 shows the kernel density estimate (KDE) and cumulative distribution function (CDF) of the execution time for the MO_case_150 algorithm in this scenario, while Figure 12 displays a violin plot of memory usage in the same scenario. This focused analysis of a single scenario helps to identify how each algorithm performs under specific workloads and reveals their potential strengths and weaknesses.

From the average time in Figure 10(a) and the average memory usage in Figure 10(b), it is evident that while the **MOGWO** algorithm performs best in terms of theabsolute time and memory usage, its running time increases significantly as the problem size grows, much more so than the other algorithms. This suggests that the time complexity of MOGWO is highly sensitive to the problem size. It excels in small to medium-sized problems but its time consumption increases rapidly with larger problem sizes. This implies that MOGWO may be better suited for small to medium-scale problems, whereas its time efficiency needs to be carefully considered for larger problems. In contrast, other algorithms such as **RLMOGWO** and **NSGAII** exhibit less pronounced increases in time, demonstrating better scalability and stability, particularly when handling large-scale problems.

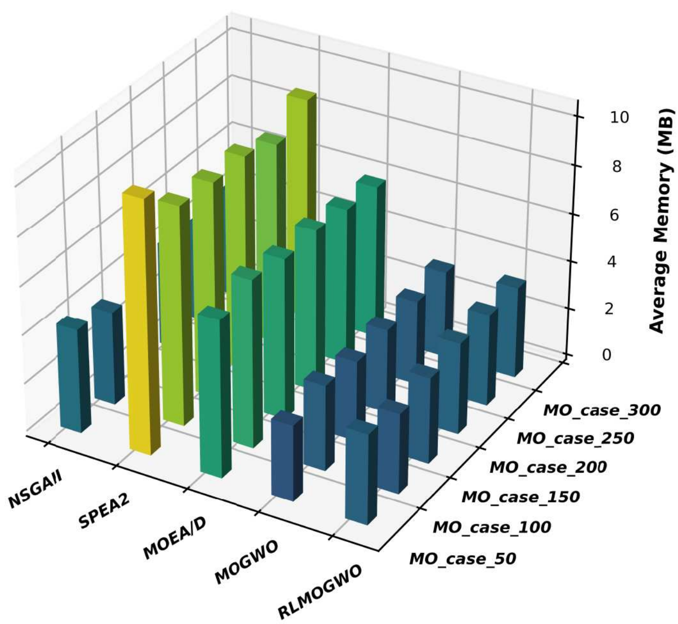
From Figure 11, in the MO_case_150 scenario, the **MOGWO** and **RLMOGWO** algorithms demonstrate the best time efficiency, with the shortest and most concentrated execution times, indicating their high performance. **NSGAII** and **MOEA/D** also show stable performance with moderate time consumption; while they are not as efficient as the former two methods, they still perform well. In contrast, **SPEA2** exhibits significantly higher time consumption with a broader distribution, making it the least efficient. Therefore, in this specific scenario, **MOGWO** and **RLMOGWO** are better suited for tasks that require high time efficiency.

As seen in Figure 12, the **MOEA/D** and **SPEA2** algorithms have the highest memory usage, with large demands and significant fluctuations, making them less suitable for situations with limited memory resources. In contrast, **MOGWO** and **RLMOGWO** exhibit lower memory consumption with more concentrated distributions, demonstrating higher memory efficiency. These algorithms are particularly well-suited for scenarios where strict control of the memory usage is required.

Based on the comprehensive analysis across various metrics, it is clear that the proposed **RLMOGWO** algorithm consistently outperforms other algorithms, particularly in scenarios with increasing complexity. RLMOGWO demonstrates a superior performance in optimizing both objective functions, minimizing IGD values, and achieving the highest HV values, all while maintaining a stable and efficient performance. Its ability to generate uniformly distributed and diverse



(a)



(b)

Figure 10. Comparison of execution time and memory usage of algorithms across scenarios. (a) Execution time of algorithms across scenarios and (b) Memory usage of algorithms across scenarios.

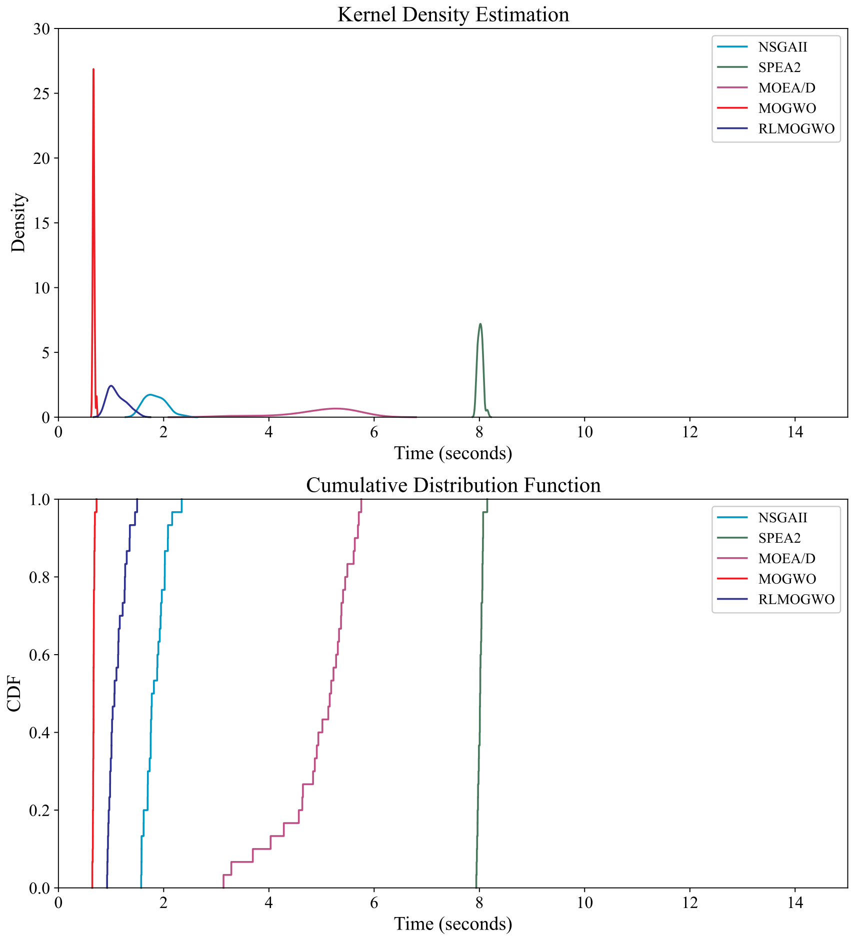


Figure 11. Kernel density estimation and cumulative distribution function of execution time for different algorithms in the MO_case_150 scenario.

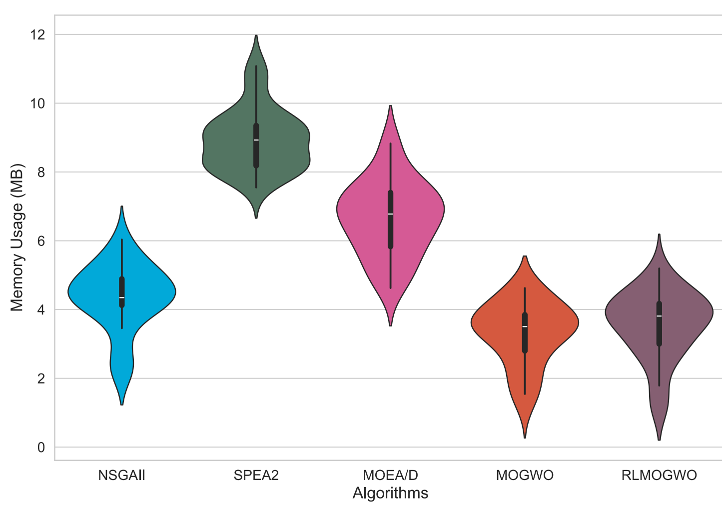


Figure 12. Violin plot of memory usage for different algorithms in the `MO_case_150` scenario.

solutions, combined with excellent scalability and resource efficiency, makes RLMOGWO the most reliable choice for multi-objective optimization tasks. Although other algorithms like **NSGAII** and **SPEA2** show a competitive performance in terms of specific aspects, RLMOGWO's overall effectiveness and dependability across different scenarios and problem sizes make it the best-suited algorithm for a wide range of optimization challenges.

5. Conclusion

This study proposes and validates a multi-objective optimization mathematical model to address the MOAEOSSP. By innovatively integrating Q-learning into an improved multi-objective grey wolf optimization algorithm, a novel optimization algorithm – RLMOGWO – was developed. The simulation results demonstrate that RLMOGWO achieves a significant balance between profit and energy consumption optimization. By incorporating the multi-objective evaluation metric IGD into the state space of Q-learning, the algorithm significantly improves the IGD metric and outperforms existing algorithms such as NSGAII, SPEA2, MOEA/D, and MOGWO across multiple evaluation criteria, showcasing its great potential in solving the MOAEOSSP problem. Although the diversity and quantity of simulation scenarios may be limited, potentially affecting the generalizability of the results, future research will expand the test scenarios and further optimize the algorithm to enhance its performance in complex environments and explore its application in other multi-objective optimization problems. The RLMOGWO algorithm holds broad application prospects in fields such as emergency response, urban planning, and resource management; itcan improve the resource allocation efficiency, reduce energy consumption, and meeting the needs of various users in multi-objective optimization and resource scheduling.

Author contributions

Conceptualization: He Wang, Weiquan Huang and Yanjie Song; Methodology: He Wang, Sindri Magnússon and Tony Lindgren; Software: He Wang ; Validation: He Wang ; Visualization: Chen Chen and Junyu Wu; Writing–original draft preparation: He Wang; Writing–review and editing: He Wang, Sindri Magnússon and Tony Lindgren,

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Data availability

The data or code used in this study are available from the corresponding author upon reasonable request.

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